

CHADI CHAHID

PhD Candidate in Applied Mathematics

Numerical Analysis of PDEs, Physics-Informed Neural Networks, and Domain Decomposition Methods

Faculty of Sciences, Moulay Ismail University, Meknès, Morocco

/chadichahid
/chahidchadi
ch.chahid@edu.umi.ac.ma
+212 674 583 393

RESEARCH PROFILE

PhD candidate in applied mathematics at Moulay Ismail University, working on numerical PDE analysis and scientific computing. My research focuses on the convergence of physics-informed neural networks (PINNs), both as standalone solvers and when combined with parallel domain decomposition methods for higher-order PDEs. I study interface transmission conditions and use domain decomposition to simplify training and reduce computational cost. I implement these methods in Python/PyTorch and assess their accuracy, robustness, and efficiency. I am interested in extending this work to the analysis and comparison of full- and reduced-order hemodynamic models, including how model parameters affect quantities of interest.

EDUCATION AND DOCTORAL RESEARCH

10/2023 – present **PhD in Applied Mathematics** — **Moulay Ismail University**

Doctoral thesis (in progress): *Convergence Analysis of Physics-Informed Neural Networks and Schwarz Domain Decomposition Methods for PDEs*

Affiliation: Faculty of Sciences, Moulay Ismail University, Meknès, Morocco

Research supervisor: Prof. Samir Khallouq

Thesis summary: My doctoral research focuses on developing and analysing space–time parallel methods for PDEs by combining domain decomposition with physics-informed neural networks. The aim is to simplify training and reduce computational cost, while establishing convergence results and error estimates for the resulting methods.

09/2021 – 09/2023 **Master's in Numerical Analysis and Partial Differential Equations** — **Moulay Ismail University**
Thesis: Schwarz-type domain decomposition methods
Supervisor: Prof. Khallouq Samir

09/2017 – 07/2021 **Bachelor's in Applied Mathematics** — **Moulay Ismail University**
Final Project: Solving differential equations using Laplace transform
Supervisor: Prof. Mohamed Zitane

07/2017 **Baccalaureate in Mathematical Sciences B** — **Moulay Ismail High School, Meknes, Morocco**

PUBLICATIONS, MANUSCRIPTS AND CURRENT RESEARCH

2026 **Published article**

Restricted additive Schwarz algorithm for a θ finite difference method to solve a two-dimensional 4th-order partial differential equation with variable coefficient

Chadi Chahid, Yassin Khali, Samir Khallouq, and Nabila Nagid.
Statistics, Optimization & Information Computing, Vol. 15, No. 6, pp. 5170–5192, 2026.
DOI: 10.19139/soic-2310-5070-3021 / Article page / Open-access PDF.
Contribution: Derivation of a priori estimates and stability and convergence analysis for a θ finite-difference discretization of a variable-coefficient fourth-order thin-plate equation. The problem is reformulated as a coupled system of second-order equations, and a restricted additive Schwarz method is employed as a preconditioner for GMRES. Numerical experiments assess the convergence and computational effectiveness of the resulting method.

2026

Manuscript under review

Order-Reduced Optimized Schwarz Physics-Informed Neural Network Method for the Euler–Bernoulli Beam Equation

Abdelhalim Ben Bouzid, **Chadi Chahid**, and Samir Khallouq.
Submitted to *Computers & Mathematics with Applications* (manuscript CAMWA-D-26-00947); currently under review.
Reproducible code and numerical experiments.
Contribution: Development of an order-reduced formulation that transforms the time-dependent fourth-order Euler–Bernoulli beam equation into a second-order complex-valued problem. The formulation is combined with optimized Schwarz transmission operators and PINN subdomain solvers. Fourier analysis is used to derive optimized transmission conditions, while implementable local and nonlocal approximations are assessed numerically in terms of convergence, accuracy, robustness, and computational cost.

2026

Manuscript in preparation

Convergence Analysis of Physics-Informed Neural Networks and a Parallel Overlapping Schwarz–PINN Method for the Euler–Bernoulli Beam Equation

Chadi Chahid and Samir Khallouq.
Current doctoral manuscript in preparation.
Contribution: Development of a unified convergence framework for PINN approximations and a parallel overlapping Schwarz–PINN method for the stationary fourth-order Euler–Bernoulli beam equation. The analysis combines sample-size-dependent convergence estimates for the local PINN solvers with contraction of the exact Schwarz interface operator to quantify the propagation of local neural approximation errors and derive global DDM–PINN error bounds. Numerical experiments in Python/PyTorch evaluate the predicted convergence behavior, interface contraction, approximation accuracy, and computational efficiency.

ACADEMIC EXPERIENCE

02/2025 – present

ENSAM – Moulay Ismail University

Teaching Assistant – Non-permanent

Teaching load: 12 hours/week.
Laboratory sessions for first- and second-year engineering students on interpolation, numerical approximation, and methods for solving linear systems.

01/2024 – 06/2024

Faculty of Sciences – Moulay Ismail University

Teaching Assistant – Non-permanent

Teaching load: 9 hours/week.
Tutorial sessions in first-year analysis, including Taylor series and asymptotic expansions; preparation and assessment of exercises; individualized academic support.

SELECTED MASTER'S COURSES

Approx. 60 h

Moulay Ismail University

Linear Partial Differential Equations

Study of linear PDEs, differential operators, and boundary-value problems. The course covered analytical tools for formulating these problems and studying their solutions.

Approx. 60 h

Moulay Ismail University

Finite Difference Methods and Scientific Computing

Construction and implementation of finite-difference schemes for differential equations. Study of their accuracy, stability, and convergence, with numerical experiments to examine the behaviour of the schemes.

Approx. 60 h

Moulay Ismail University

Finite Element Methods and Simulation

Introduction to finite-element methods for PDEs, including the construction of discrete approximations. Application of these methods to numerical simulation of PDE problems.

SCIENTIFIC COMMUNICATIONS

20–21 May 2025 — **JIAMA'25 – ENSA Al Hoceima**

Convergence Analysis of PINNs for Fourth-Order PDEs

Third International Conference on Artificial Intelligence and Applied Mathematics, Al Hoceima, Morocco.

9–10 July 2025 — **MISI'25 – ENSA Safi**

Error Analysis of PINNs Approximating the Euler–Bernoulli Beam Equation

International Conference on Mathematics and Intelligent Systems in Industry, Safi, Morocco.

1–3 December 2025 — **SM2A 2025 – Meknes**

Non-overlapping Schwarz Domain Decomposition Coupled with Physics-Informed Neural Networks (PINNs) to Solve the Euler–Bernoulli Beam Equation

Oral communication in the Numerical Analysis session of the 7th International Congress of the Moroccan Society of Applied Mathematics, Faculty of Sciences, Moulay Ismail University, Meknes, Morocco. Joint work with Abdelhalim Ben Bouzid and Samir Khallouq.

SCIENTIFIC AND COMPUTATIONAL EXPERTISE

Numerical Methods

Finite-difference methods; overlapping and restricted additive Schwarz methods; numerical approximation of PDEs; stability, convergence, and error analysis.

Scientific Machine Learning

Physics-informed neural networks; automatic differentiation; Hölder/Lipschitz regularization; Adam and L-BFGS optimization.

Programming and Scientific Computing

Python; PyTorch; NumPy; SciPy; Matplotlib; scikit-learn; Git; Google Colab; LaTeX.

SELECTED APPLIED PROJECT

03/2024 – 05/2024 — **ALM-Solutions collaboration**

Credit Card Fraud Detection System

Data preparation, classification under class imbalance, model evaluation using precision, recall, and F1 score, and integration of a real-time fraud detection prototype using Python and scikit-learn.

LANGUAGES

Arabic	Native proficiency
French	Full professional
English	Full professional
Dutch	Elementary

SCIENTIFIC TOOLS

Python / PyTorch
PINNs, automatic differentiation, optimization

NumPy / SciPy / Matplotlib
Numerical computation and visualization

Git / Google Colab / LaTeX
Reproducible computing and scientific writing

ACADEMIC STRENGTHS

Rigorous mathematical analysis
Convergence, stability, and approximation estimates

Research communication
Mathematical writing and conference presentations

Collaborative research
Numerical implementation and reproducible experiments

Personal website: www.chahidchadi.github.io